

Stellenbosch University Faculty of Engineering

Module Framework

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This document should be read with the following documents:

- Stellenbosch University Calendar Parts 1 and 11.
- Faculty of Engineering Assessment Rules¹
- Faculty of Engineering General Stipulations for Undergraduate Modules¹

Project (E) 448 46795-448 2026, Sem 2	Lecturer(s): Module coordinator: Prof MM Botha, Room E2048, mmbbotha@sun.ac.za Module lecturers: All E&E Eng academic staff employed during 2026. Internal moderator: Prof A Rix	Approved by Programme Coordinator:  Date: 7/2/2026
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1. Assessment Details

- Major assessment dates and venues are provided at <https://web-apps.sun.ac.za/academic-exam-timetable/#/start/>
- Method of assessment as indicated in the Calendar Part 11
- Note that awarding a pass mark is subject to meeting each ECSA Graduate Attribute assessed in this module, as stated in Faculty of Engineering's Assessment Rule

Calculation of final marks (according to formulas in the Faculty of Engineering's Assessment Rules):

Project assessment: final mark awarded after evaluation of project, presentation and poster session.

2. Notional Hours

- You should spend 10 notional hours per credit on this module over the course of the semester.
- It is envisaged that these hours will be allocated as follows:

Activity	Contact Hours	Self-study Hours
Lectures		
Tutorials		
Practicals		
Assignments	20	430
Main assessments		
Total (for this 45-credit module)	450	

3. Language of Tuition

- The language of tuition in this module is according to the Faculty's approved Language Implementation Plan. Please refer to the website of the Engineering Faculty or the "General Information" block on STEMLearn for the particulars.

4. Module Objectives

Aim: To complete an individual project involving project planning, problem identification, problem solving, design, implementation, evaluation and documentation.

A student who has successfully completed this module can/This module develops students' abilities to:

- Take responsibility for planning a project and for independent learning

- Identify engineering problems and solve them
- Apply scientific and engineering knowledge
- Design engineering solutions
- Design and conduct investigations and experiments and analyse results
- Use appropriate engineering methods, skills and tools
- Document and communicate methods and results

See the listed ECSA GAs for further details.

¹ Available on STEMLearn for modules offered by Faculty of Engineering, in the block titled "General Programme Information" on the side of the screen

5. Module Content and Schedule

Prescribed textbook(s): NA		
Semester 1		
Week	Topic	Contact Session/Assignments
All semester weeks	Project work.	Each student must regularly meet with their project study leader.
Jun 2, 12:00	Project report hand in deadline. Hand in electronically via STEMLearn. Penalty for late submission is 5% per half day.	N/A
Jun 8-9	Oral presentations and internal examination in this period.	N/A
Jun 8, 23:59	Deadline to voluntarily upload a PDF version of the oral presentation slides. These slides will only be relevant for "Case 2" assessments. In case of failure to upload the slides, the Case 2 option falls away.	N/A
Jun 15	Interviews with external moderators. You must be available on this date to present your poster. This is a requirement for passing the module.	N/A
Semester 2		
Week	Topic	Contact Session/Assignments
All semester weeks	Project work.	Each student must regularly meet with their project study leader.
Oct 30, 23:59	Project report hand in deadline. Hand in electronically via STEMLearn. Penalty for late submission is 5% per half day.	N/A
Nov 6-13	Oral presentations and internal examination in this period.	N/A
Nov 11, 23:59	Deadline to voluntarily upload a PDF version of the oral presentation slides. These slides will only be relevant for "Case 2" assessments. In case of failure to upload the slides, the Case 2 option falls away.	N/A
Nov 18	Project open day. Interviews with external moderators. You must be in attendance and present your poster. This is a requirement for passing the module.	N/A

The nominal dates/deadlines above may be changed by the lecturer if necessary, given public holidays, etc. **Always see STEMLearn for the latest information.**

6. ECSA Knowledge Area Credits

Mathematical Sciences	Natural Sciences	Engineering Sciences	Design and Synthesis	Complementary Studies
0	0	0	38	7
<u>Design and Synthesis:</u> A project involving design and synthesis must be completed.				
<u>Complementary Studies:</u> The work involves project planning, documentation and presentation.				

7. ECSA Graduate Attributes (GA)

Students are required to complete individual projects, each supervised by a staff member. The problems are varied and contain convergent and divergent aspects, and are of a higher level of complexity than that of Design 314. No structured problem solving method is enforced in a group context.

GA 1. Problem solving

How is the Outcome Assessed?	Assessment is based on a written project report and an oral examination (for Cases 1 and 2), possibly also accompanied by the student's oral presentation slides (only for Case 2). In the project report, the student is required to indicate how they have achieved this outcome by referring to the relevant parts of their report. The oral examination is carried out by two internal examiners under supervision of a convener. The two internal examiners and one external moderator must indicate explicitly on the assessment forms that the student has complied with the required outcome. • Case 1: The project report provides the evidence. The internal oral examination's contribution to this assessment is only to aid the two internal examiners' understanding of the evidence on record (i.e. the report). • Case 2: The GA is not achieved with the report, but the presentation slides provide sufficient additional documented evidence for its achievement. A copy of the slides must be attached to the report. A maximum mark of 50% applies. Also see the assessment forms.
What is Satisfactory Performance?	Using the assessment opportunities, the student must show that they satisfy this GA at a level consistent to that at which a graduate would participate within an employment situation shortly after graduation. Complex engineering problems: (a) These require a fundamentals-based, first principles analytical approach, use in-depth engineering knowledge and have one or more of the following characteristics: - A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline. - Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much of which is at the forefront of the discipline. - Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area. - Knowledge of engineering practice (technology) in the practice areas in the engineering discipline. - Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues. (b) and have some or all of the following characteristics: - Involve wide-ranging and/or conflicting technical, non-technical issues (such as ethical, sustainability, legal, political, economic, societal) and consideration of future requirements. - Have no obvious solution and require abstract thinking, creativity and originality in analysis to formulate suitable models. - Involve infrequently encountered issues or novel problems. - Address problems not encompassed by standards and codes of practice for professional engineering. - Involve collaboration across engineering disciplines, other fields, and/or diverse groups of stakeholders with widely varying needs. - Address high-level problems with many components or sub-problems that may require a systems approach. This GA entails the following: Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development. Associated knowledge and attitude profile: • A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences. • Conceptually based mathematics, numerical analysis, data analysis, statistics and formal aspects of

	computer and information science to support detailed analysis and modelling applicable to the discipline. • A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline. • Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline, much of which is at the forefront of the discipline.
What is the consequence of unsatisfactory performance?	If the candidate has not achieved the outcome, he/she cannot pass.

GA 2. Application of scientific and engineering knowledge

How is the Outcome Assessed?	In the same way as GA 1 (see the GA 1 assessment description).
What is Satisfactory Performance?	Using the assessment opportunities, the student must show that they satisfy this GA at a level consistent to that at which a graduate would participate within an employment situation shortly after graduation. This GA entails the following. Definition: Apply knowledge of mathematics, natural science, computing and engineering fundamentals, and an engineering specialisation to develop solutions to complex engineering problems. Associated knowledge and attitude profile: • As for GA1. Range statement: Mathematics, natural science and engineering sciences are applied in formal analysis and modelling of engineering situations, and for reasoning about and conceptualising engineering problems.
What is the consequence of unsatisfactory performance?	If the candidate has not achieved the outcome, he/she cannot pass.

GA 3. Engineering Design

How is the Outcome Assessed?	In the same way as GA 1 (see the GA 1 assessment description).
What is Satisfactory Performance?	Using the assessment opportunities, the student must show that they satisfy this GA at a level consistent to that at which a graduate would participate within an employment situation shortly after graduation. This GA entails the following. Definition: Design creative solutions for complex engineering problems and design systems, components or processes to meet identified needs. Associated knowledge and attitude profile: Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon and similar concepts, that supports engineering design and operations in a practice area. Range statement: Design problems used in exit-level assessment must conform to the definition of a complex engineering problem. A major design problem should be used to provide evidence. The design knowledge base and components, systems, engineering works, products or processes to be designed are dependent on the discipline or practice area. Appropriate consideration must be given to public health and safety, whole-life cost and net zero carbon, as well as resource, cultural, societal and environmental considerations, as required.

What is the consequence of unsatisfactory performance?	If the candidate has not achieved the outcome, he/she cannot pass.
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GA 4. Investigations, experiments and data analysis

How is the outcome Assessed?	In the same way as GA 1 (see the GA 1 assessment description).
What is Satisfactory Performance?	Using the assessment opportunities, the student must show that they satisfy this GA at a level consistent to that at which a graduate would participate within an employment situation shortly after graduation. This GA entails the following. Definition: Demonstrate competence to conduct investigations of complex engineering problems using research methods, including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions. Associated knowledge and attitude profile: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues. Range statement: The balance of investigation and experiment should be appropriate to the discipline. Research methodology is to be applied in research or an investigation where the student engages with selected knowledge in the research literature of the discipline. Note: An investigation differs from a design in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artefact can be produced.
What is the consequence of unsatisfactory performance?	If the candidate has not achieved the outcome, he/she cannot pass.

GA 5. Use of engineering tools

How is the Outcome Assessed?	In the same way as GA 1 (see the GA 1 assessment description).
What is Satisfactory Performance?	Using the assessment opportunities, the student must show that they satisfy this GA at a level consistent to that at which a graduate would participate within an employment situation shortly after graduation. This GA entails the following. Definition: Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems. Associated knowledge and attitude profile: • Concept based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science in support of detailed analysis and modelling, applicable to the discipline • Knowledge of the engineering practices (technology) in the practice of the engineering discipline Range statement: A range of techniques, resources and modern engineering and IT tools appropriate to the disciplinary designation of the programme.

What is the consequence of unsatisfactory performance?	If the candidate has not achieved the outcome, he/she cannot pass.
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GA 6. Professional and technical communication

How is the Outcome Assessed?	Assessment is based on a written project report, an oral presentation and a poster presentation (the latter is open to the general public). The oral presentation is assessed by the two internal examiners and the poster presentation is assessed by the external moderator. The two internal examiners and one external moderator must indicate explicitly on the assessment forms that the student has complied with the required outcome. Also see the assessment forms.
What is Satisfactory Performance?	Using the assessment opportunities, the student must show that they satisfy this GA at a level consistent to that at which a graduate would participate within an employment situation shortly after graduation. This GA entails the following. Definition: Demonstrate competence to communicate effectively and inclusively on complex engineering activities, both orally and in writing, with the engineering community and society at large, taking into account cultural, language, and learning differences. Range statement: Material to be communicated is in an academic or simulated professional context. Audiences range from engineering peers, management and lay persons, using appropriate academic or professional discourse. The long written report (10 000 to 15 000 words plus tables, diagrams and appendices), covers material at exit-level. Methods of providing information include the conventional methods of the discipline, for example engineering drawings and design documentation, as well as subject-specific methods. The poster must be accessible to lay persons, using appropriate academic or professional discourse, while taking into account cultural, language, and learning differences.
What is the consequence of unsatisfactory performance?	If the candidate has not achieved the outcome, he/she cannot pass.

GA 8. Individual work

How is the Outcome Assessed?	In the same way as GA 1 (see the GA 1 assessment description).
What is Satisfactory Performance?	Using the assessment opportunities, the student must show that they satisfy this GA at a level consistent to that at which a graduate would participate within an employment situation shortly after graduation. This GA entails the following. Definition: Demonstrate competence to function effectively as an individual. Associated knowledge and attitude profile: Knowledge of professional ethics, responsibilities and norms of engineering practice. Range statement: Multi-disciplinary tasks require co-operation across at least one disciplinary boundary. Co-operating disciplines may be engineering disciplines with different fundamental bases other than that of the programme or may be outside engineering. Developmental considerations: • Develop the ability to self-reflect and self-evaluate within an interpersonal engagement towards enabling appropriate understanding of self. • Time management also includes understanding the value of time, and determining if a task is better (cheaper) achieved by a single person or a team.

What is the consequence of unsatisfactory performance?	If the candidate has not achieved the outcome, he/she cannot pass.
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GA 9. Independent learning ability

How is the Outcome Assessed?	In the same way as GA 1 (see the GA 1 assessment description).
What is Satisfactory Performance?	Using the assessment opportunities, the student must show that they satisfy this GA at a level consistent to that at which a graduate would participate within an employment situation shortly after graduation. This GA entails the following. Definition: Demonstrate competence to engage in independent learning through well-developed learning skills. Associated knowledge and attitude profile: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues. Range statement: Operate independently in complex contexts recognising the need for and having the preparation and ability for i) independent and life-long learning, ii) adaptability to new and emerging technologies, and iii) critical thinking in the broadest context of technological change. Developmental considerations: • Openness to constructive feedback, awareness of own limitations, ability to cope with the discomfort of uncertainty and having access to a range of approaches, reflective self-evaluation, curiosity and proactive engagement, resilience, confidence to ask for help and draw from a broad range of stakeholders. • Reflection of self-learning to begin to recognise if what has been covered meets the needs of the activity or task.
What is the consequence of unsatisfactory performance?	If the candidate has not achieved the outcome, he/she cannot pass.

8. Other Module Specific Information

8.1. Responsibilities and Student's GA Achievement Plan

The candidate makes an appointment with the project leader as soon as possible after the announcement of the project allocations. At this meeting, the aims, methods, requirements and planning of the project are discussed. The final project proposal is summarized in the form of the agreement (attached and also made available via the module website). This agreement between the project leader and the candidate clearly states the expectations and responsibilities of both parties. The student must also explicitly explain in this document how they aim to achieve **all** the required the GAs. The agreement must be completed by the end of the 4th week of the second semester and submitted via STEMLearn.

The project leader and the candidate must agree on a time for their weekly appointments.

The project study leader must give guidance to the candidate by:

- Making the candidate aware of the departmental policy with regard to the course module;
- Honouring the weekly appointments with the candidate;
- Verifying on a continuous basis that the candidate has clarity with regard to the aims of the project, and that the candidate's planning for the project is a true reflection of the aims;
- Monitoring the progress of the project, and encouraging the student to have a critical approach to the problem;
- Spelling out the consequences of plagiarism;
- Explanation of the importance of effective communication;
- Emphasising the importance of verification that all the outcomes were satisfied;
- Giving feedback at least once, with regards to a draft version of the report.

During the execution of the project the candidate must:

- Adhere to the policy with regard to the course module, and ensure that all due dates are honoured;
- Attend the weekly appointments with the project leader;
- With the exception of field measurements and report writing, work in the buildings of the Faculty as much as possible;
- Verify that all the ECSA GAs are satisfied;
- Compile a planning schedule (Appendix A in project report) for the project. The candidate must present a preliminary planning schedule by the second week of the semester to their project leader.

8.2. Evaluation

The final mark is determined by evaluation of the following:

- Project report
- Oral presentation
- Poster presentation

Each project is internally evaluated by two examiners and then externally moderated. The two internal examiners determine whether all the claimed outcomes are achieved and award a preliminary, consensus mark. In case no fail/pass consensus is achieved by the two internal examiners, a third internal examiner is appointed.

The external moderator is responsible for the final verification of outcomes and determination of the final mark. They have access to the examination reports of the internal examiners, as well as the project reports (and possibly the presentation slides in special cases --- see “Case 2” below) and the poster presentation. The final marks are verified by the external moderator with their signature. That is the final certification of achievement of claimed outcomes. Penalties for late submission are subtracted from these final marks. The marks are then verified by the internal module moderator, as appointed by the Departmental Executive.

Notes on Case 2:

- ***“Case 2” above provides the student with the opportunity to extend/finish their work during the period between report hand in and the oral, in case critical content is absent from the report such that they might likely fail the “Case 1” assessment.***
- ***Case 2 limits the mark to $\leq 50\%$.***
- ***It is recommended that the student inform their project leader ahead of the oral, of their intention to present significant further work at the oral through comprehensive documentation via their slides, with an aim towards achieving outcomes which otherwise might likely not be achieved.***
- ***Additional work beyond the contents of the report, which is presented at the oral, will not be taken into account for a Case 1 assessment.***
- ***Presenting additional work at the oral does not automatically place the student in the Case 2 category.***
- ***Case 2 is only engaged when it is to the benefit of the student.***
- ***It is the student’s responsibility to upload their slides to the module website by the deadline indicated. Students will not be informed that they will be regarded as Case 2. Failure to upload slides will lead to the assessment reverting back to Case 1.***

The evaluation forms are attached.

8.3. Project report

An electronic copy (PDF) of the project report must be submitted via the module website by the due date and time as specified in the module schedule above. TurnItIn will be used to detect plagiarism.

Please format the file name of your uploaded file as follows:

(studentnumber)_(surname)_(initials)_skripsie(year).pdf

Example: 1234567_Bloggs_PJ_skripsie2015.pdf

The required report format is as follows:

- Typeset on A4 paper with 12-point font.
- Maximum length of the body of the report: 40 pages (the body starts on the first page of the Introduction and ends on the last page of the Conclusion).
 - 10% may be deducted from the final mark for an over-length report.
- All tables, graphs, diagrams and photos must be numbered and have captions.
- Margins: left 2.5 cm, right 2.5 cm, top 2.5 cm, bottom 2.5 cm.

The required report sections are:

- Title page, with the following:
 - Project title
 - Name of candidate.
 - Student number of the candidate.
 - The words "Report submitted in partial fulfilment of the requirements of the module Project (E) 448 for the degree Baccalaureus in Engineering in the Department of Electrical and Electronic Engineering at the University of Stellenbosch".
 - STUDY LEADER: (Name of lecturer(s));
 - DATE: (Month and Year).
- Page with acknowledgements.
- A page with the official SU plagiarism declaration, signed.
- Page with summary(ies) in Afrikaans and/or English, each maximum of 150 words.
- Table of Contents.
- Lists with figures, tables and symbols.
- Chapter 1: Introduction, where the background to the project and the project aims are described. A short summary of the report can also be given.
- Chapters 2..N-1: Body of the report, and results. Be precise and concise.
- Chapter N: Conclusions and recommendations.
- Literature references using IEEE or Harvard format.
- Appendix A: Project planning schedule.
- **Appendix B: Outcomes compliance, which should have a section for each GA, explaining how it is achieved during the execution of the project, with reference to the relevant report sections. Note the following particular points.**
 - GA1: explain how sustainable development is considered in your problem and solution, or envisaged extensions of it.
 - GA3 requires that consideration be given to public health and safety, whole-life cost and net zero carbon, as well as resource, cultural, societal

and environmental considerations. If these aspects did not feature explicitly in the project work, then this section should discuss how these aspects could relate to the project work or envisaged extensions of it.

- GA4 and GA9: you have to reflect on how you applied critical thinking.
 - GA4 and GA9: you have to explain how you evaluated emerging issues in your report, or do that in these sections, in relation to the project work or envisaged extensions of it.
 - GA9: you have to explain how you evaluated emerging technologies in your report, or do that in this section, in relation to the project work or envisaged extensions of it.
 - GA1 and GA2: if awareness of relevant social sciences were required for your project, then explain how and where this features.
 - GA5: you should explain how creating, selecting, and application of engineering tools feature in your project, or could feature for certain aspects in envisaged extensions of it. You should also explain where you recognised the limitations of tools/technologies in your project, or can recognize this in envisaged extensions of it.
 - GA6: remember that the poster must be accessible to lay persons, using appropriate academic or professional discourse, while taking into account cultural, language, and learning differences.
 - GA9: discuss any an all of these aspects, that did feature during the course of your project work: “Openness to constructive feedback, awareness of own limitations, ability to cope with the discomfort of uncertainty and having access to a range of approaches, reflective self-evaluation, curiosity and proactive engagement, resilience, confidence to ask for help and draw from a broad range of stakeholders.
 - GA9: Provide a reflection on self-learning to begin to recognise if what has been covered meets the needs of the project work.
- Appendices, that may include circuit diagrams, measured data, derivations, programs, printouts, etc. Note examiners will mostly look at the main body in evaluating your work.

8.4. Oral examination

- The oral examination is used in conjunction with the main report to determine the extent to which the candidate was able to satisfy the outcomes of the module.
- The module coordinator appoints two internal examiners, consisting of the project leader and another lecturer, as well as a convener for the oral. The module coordinator gives a copy of the project report to each of the examiners.
- The format of the oral examination is as follows:
- The evaluation is chaired by the convener of the oral.
- The candidate has 10 minutes for their oral presentation. This might be via an uploaded video or virtual/in-person presentation. For in-person presentation the use of the provided projector is advised; organize a laptop yourself (ask your project leader if needed).
- The two internal examiners can ask questions to determine the extent to which the candidate has mastered the work and the candidate is excused.
- The study leader gives an indication of the extent of the guidance that was required by the candidate.
- The preliminary reports from the two internal examiners, with their preliminary mark allocation, are handed to the convener. The evaluations are based on the evaluation form and ensure that all the required outcomes were considered in awarding the preliminary mark.
- The consensus outcomes assessment and recommended final mark are determined after discussion. The following recommendations are possible (subject to the verification of the external moderator):
- The candidate passes with $PP \geq 50\%$ upon evidence of the report.
- The candidate passes with $PP = 50\%$, upon evidence of both the report and the presentation slides.
- The candidate fails with $PP < 50\%$. The course module must be repeated with a different topic. Note that there is no option for improvement of the work/report beyond this point.
- In the case where the examiners cannot reach pass/fail consensus, a third internal examiner is appointed by the module coordinator.
- Afterwards, it is the responsibility of the Convenor to hand over the two preliminary evaluation forms and the convener report form containing the recommended final mark, to the project coordinator, via a departmental administrative officer.

8.5. Open Day and poster presentation

The external moderators conduct their moderation work directly after completion of the internal examination process. The Project Open Day is scheduled to coincide with the completion of their work. Every candidate must prepare a poster that is presented at the Project Open Day. The poster (A1 size) must give an overview of the scope of the project, as well as the main results (also see GA6 requirements on the poster). The candidate must also prepare a short oral presentation (3 minutes maximum) to give the external moderators an overview of the project, if so requested.

The final verification of the claimed outcomes and mark cannot take place in the absence of the candidates, and attendance of the Project Open Day is therefore compulsory. Non-attendance will lead to an incomplete assessment for the module.

8.6. A note on project assignment

This happens during the first semester. Individual preference data collection from students is accomplished via the module website and a cut-off date for students to submit their preferences is communicated via the website. Based on the preference data, students are assigned to supervisors. The assignments are made known before the end of the examination period of the first semester. Students are strongly advised to make an appointment with their project supervisors before the start of the July vacation in order to be able to make preliminary preparations during the vacation period.

8.7. Communication with students

STEMLearn will be used as the primary channel of communication when disseminating information about this module.

8.8. Forms

The project agreement form and assessment forms are attached.

Each examiner must complete this form independently, and then hand it over to the convener.

Student

Initials and surname		SU number	
Project title			

Internal examiner

Initials and surname	Signature	Role (<i>tick one</i>)
		☐ Supervisor ☐ 2nd examiner

Grading assessment

Only for Case 2: Tick if the oral presentation slides provide necessary evidence in order to achieve certain GAs, and list those GA numbers (in this case a mark of $\leq 50\%$ must be awarded)	
☐ GA(s): _____	
Preliminary mark • Provide a percentage, or percentage range • Failed GA(s) $\Rightarrow \leq 45\%$	
Comments in support of the preliminary assessment:	

Graduate attribute (GA) assessment

Recorded evidence requirements, for GA achievement at the level expected of a recent graduate	Tick one in each row			
	Student satisfies the GA criteria, taken as a whole		Student fails to satisfy the GA criteria, taken as a whole	
	Satisfied	Marginally satisfied	Marginally not satisfied	Not satisfied
GAs 1–5, 8 and 9: Case 1: The project report provides the evidence. The internal oral exam's contribution is only to aid internal examiners' understanding of the evidence on record (i.e. the report). Case 2: Some GAs are not achieved with the report, but the presentation slides provide evidence for their achievement. These GAs may be marked as achieved, provided that a copy of the slides is attached to the report. A maximum mark of 50% applies. GA 6: The project report (written communication) AND a satisfactory oral (oral communication)				
GA 1: Problem solving (Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development.) See module framework for full description	☐	☐	☐	☐
GA 2: Application of scientific and engineering knowledge (Apply knowledge of mathematics, natural science, computing and engineering fundamentals, and an engineering specialisation to develop solutions to complex engineering problems.) See module framework for full description	☐	☐	☐	☐
GA 3: Engineering design (Design creative solutions for complex engineering problems and design systems, components or processes to meet identified needs.) See module framework for full description	☐	☐	☐	☐
GA 4: Investigations, experiments and data analysis (Demonstrate competence to conduct investigations of complex engineering problems using research methods, including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.) See module framework for full description	☐	☐	☐	☐
GA 5: Use of engineering tools (Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.) See module framework for full description	☐	☐	☐	☐
GA 6: Professional and technical communication (Demonstrate competence to communicate effectively and inclusively on complex engineering activities, both orally and in writing, with the engineering community and society at large, taking into account cultural, language, and learning differences.) See module framework for full description	☐	☐	☐	☐
GA 8: Individual work (Demonstrate competence to function effectively as an individual.) See module framework for full description	☐	☐	☐	☐
GA 9: Independent learning ability (Demonstrate competence to engage in independent learning through well-developed learning skills.) See module framework for full description	☐	☐	☐	☐

After receiving the two examiner reports the convener facilitates a discussion between the two examiners, with the aim of establishing a consensus assessment, which is documented here.
 In case consensus GA succeed/fail assessment(s) differ from one/both examiner's report(s), or no consensus occurs, then the convener must document the examination panel's motivations.
 The convener returns the completed convener and two examiner reports to the module administrator.

Student

Initials and surname		SU number	
Project title			

Internal examination panel

Role	Initials and surname	Signature	
Supervisor (and 1st examiner)			
2nd examiner			
Convener			
Date		Time	

Grading consensus assessment

Fill only for Case 2 slide-based GA achievement **and/or** failed GAs **and/or** failure to achieve consensus

☐ **Case 2:** Tick and list if GA(s) are achieved by consensus, on **presentation slide evidence**.

GA(s): _____

☐ Tick in case **consensus was reached** that one or more GAs are **not** achieved.

☐ Tick in case **consensus was not reached** on one or more GAs.

☐ Tick in case **consensus was not reached** on the mark.

Recommended mark

- In case of slide-based GA evidence, ≤50% must be awarded
- In case of consensus on ANY failed GA(s), ≤45% must be awarded
- In case of NO mark consensus, provide both internal examiner marks (no ranges, supervisor first, 2nd examiner second)

Comments in support of the recommended mark –OR– Reason(s) for no mark consensus (only needed if these are different from the provided reasons for GA non-consensus):

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Graduate attribute (GA) consensus assessment

Recorded evidence requirements, for GA achievement at the level expected of a recent graduate	Tick one in each row				
		Student satisfies the GA criteria, taken as a whole		Student fails to satisfy the GA criteria, taken as a whole	
GAs 1–5, 8 and 9: Case 1: The project report provides the evidence. The internal oral exam's contribution is only to aid internal examiners' understanding of the evidence on record (i.e. the report). Case 2: Some GAs are not achieved with the report, but the presentation slides provide evidence for their achievement. These GAs may be marked as achieved, provided that a copy of the slides is attached to the report. A maximum mark of 50% applies. GA 6: The project report (written communication) AND a satisfactory oral (oral communication)	No consensus	Satisfied	Marginally satisfied	Marginally not satisfied	Not satisfied
GA 1: Problem solving • See module framework for full description	☐	☐	☐	☐	☐
GA 2: Application of scientific and engineering knowledge • See module framework for full description	☐	☐	☐	☐	☐
GA 3: Engineering design • See module framework for full description	☐	☐	☐	☐	☐
GA 4: Investigations, experiments and data analysis • See module framework for full description	☐	☐	☐	☐	☐
GA 5: Engineering methods, skills and tools, including information technology • See module framework for full description	☐	☐	☐	☐	☐
GA 6: Professional and technical communication • See module framework for full description	☐	☐	☐	☐	☐
GA 8: Individual work • See module framework for full description	☐	☐	☐	☐	☐
GA 9: Independent learning ability • See module framework for full description	☐	☐	☐	☐	☐
Motivation for every GA where succeed/fail consensus differs from examiner report(s), or with no consensus:					

The external moderator makes an assessment of the work, based upon the internal examination panel's reports, the project report (possibly with attached oral presentation slides) and the poster.

Student

Initials and surname		SU number	
Project title			

External moderator

Initials and surname	Signature	Date

Grading assessment

Are all GAs achieved? (<i>tick one</i>)		☐ Yes	☐ No
Internal, recommended consensus mark (from convener form)		External, final mark • <i>Slide-based GA evidence</i> $\Rightarrow \leq 50\%$ • <i>Failed GA(s)</i> $\Rightarrow \leq 45\%$	
Comments:			

Graduate attribute (GA) assessment

Recorded evidence requirements, for GA achievement at the level expected of a recent graduate	Tick one in each row	
	Student satisfies the GA criteria, taken as a whole	Student fails to satisfy the GA criteria, taken as a whole
• GAs 1–5, 8 and 9: ○ Case 1: The project report provides the evidence. The external poster session's contribution is only to aid external moderators' understanding of the evidence on record (i.e. the report). ○ Case 2: Some GAs are not achieved with the report, but the presentation slides provide evidence for their achievement. These GAs may be marked as achieved, provided that a copy of the slides is attached to the report. A maximum mark of 50% applies. • GA 6: The project report (written communication) AND satisfactory internal oral + satisfactory external poster (oral communication)		
GA 1: Problem solving (Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development.) • See module framework for full description	☐	☐
GA 2: Application of scientific and engineering knowledge (Apply knowledge of mathematics, natural science, computing and engineering fundamentals, and an engineering specialisation to develop solutions to complex engineering problems.) • See module framework for full description	☐	☐
GA 3: Engineering design (Design creative solutions for complex engineering problems and design systems, components or processes to meet identified needs.) • See module framework for full description	☐	☐
GA 4: Investigations, experiments and data analysis (Demonstrate competence to conduct investigations of complex engineering problems using research methods, including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.) • See module framework for full description	☐	☐
GA 5: Use of engineering tools (Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.) • See module framework for full description	☐	☐
GA 6: Professional and technical communication (Demonstrate competence to communicate effectively and inclusively on complex engineering activities, both orally and in writing, with the engineering community and society at large, taking into account cultural, language, and learning differences.) • See module framework for full description	☐	☐
GA 8: Individual work (Demonstrate competence to function effectively as an individual.) See module framework for full description	☐	☐
GA 9: Independent learning ability (Demonstrate competence to engage in independent learning through well-developed learning skills.) • See module framework for full description	☐	☐

Project (E) 448 Responsibilities & Student's GA Achievement Plan Ver2.2

*This form must be filled out by the student, signed and handed in to the module administrator early in the semester. Both the student and the supervisor must sign the agreement on main details and mutual responsibilities. The student should discuss their GA achievement plan with the supervisor, before completing this form. Only the student signs their GA achievement plan, as it is their own responsibility. **This plan is NOT a guaranteed recipe for passing Project (E) 448.** Rather, it serves as a record of the student having considered these important aspects at an appropriately early stage. GA achievement plans should be revised as needed and in consultation with the supervisor, during the course of the project.*

Main details

Student	Initials and surname		SU number	
Supervisor	Initials and surname			
Project title				
Project description, including the aim, scope and envisioned approach (max. 150 words)				

Mutual responsibilities

1. It is the responsibility of the student to clarify aspects such as the definition and scope of the project, the place of study, research methodology, reporting opportunities and -methods (e.g. progress reports, internal presentations and conferences) with the supervisor.
2. It is the responsibility of the supervisor to give regular guidance and feedback with regard to the literature, methodology and progress.
3. The rules regarding submission and evaluation of the project is outlined in the module framework and SUNLearn page and will be strictly adhered to.
4. The supervisor conveyed the departmental view on plagiarism to the student, and the student acknowledges the seriousness of such an offence.
5. The supervisor certifies that the project as described above has sufficient scope to achieve, in principle, the required GAs.
6. It is the responsibility of the student to initiate a discussion with the supervisor on GA achievement prior to filling out and handing in this form.

Signatures for agreement on main details and mutual responsibilities

Role	Signature	Date
Student		
Supervisor		

Student's graduate attribute (GA) achievement plan

How will GA 1 (problem solving) be achieved? (<i><=100 words</i>)
How will GA 2 (application of scientific and engineering knowledge) be achieved? (<i><=100 words</i>)
How will GA 3 (engineering design) be achieved? (<i><=100 words</i>)
How will GA 4 (investigations, experiments and data analysis) be achieved? (<i><=100 words</i>)
How will GA 5 (engineering methods, skills and tools, including IT) be achieved? (<i><=100 words</i>)
How will GA 6 (professional and technical communication) be achieved? (<i><=100 words</i>)
The project includes a written report and an oral presentation. These demonstrate competence to communicate effectively, both orally and in writing.
How will GA 8 (individual work) be achieved? (<i><=100 words</i>)
The student will take primary responsibility for successful completion of all aspects of the project.
How will GA 9 (independent learning ability) be achieved? (<i><=100 words</i>)
For successful completion of the project, the student is required to acquire knowledge independently (from the literature or the internet, for example) and without the context of this required knowledge being fully specified in the project definition.

Signature acknowledging own responsibility to achieve GAs

	Signature	Date
Student		